Nutrient cycles Feed the soil

Make compost using a wide range of ingredients and grow peas and beans to capture nitrogen from the air. Return useful material to the soil, then rotate crops to make good use of it.

All crops need three core chemical nutrients, plus a range of minerals, mostly in trace amounts.

Nutrient	What it does	Capture, store and use it
N **Nitrogen**	Leaf and shoot growth: kale, cabbage, spinach and other leafy greens.	Leave pea and bean roots in the soil after cropping. Compost the tops and follow with leafy greens.
P **Phosphorus**	Root development and energy transfer: all crops, including young transplants.	Recycle plant waste through compost. Outside sources include commercial compost from council waste schemes; rotted farm manure from organic pasture; composted tree leaves from parks and municipal gardens.
K **Potassium**	Water balance and crop quality: tomatoes, cucumbers and potatoes.	Grow comfrey in a permanent patch and add the leaves to your compost. Add ash from a woodburner or garden fire (untreated wood only) to compost, or to bare soil in winter. Avoid ash on alkaline soil or potato beds. Also gather seaweed (see below), or use a commercial seaweed extract.
Micronutrients	Healthy growth in all crops; iron and boron are needed in tiny amounts.	Most soils already have enough. Return trace nutrients by using varied compost ingredients and add seaweed if you can get it (see below).



1 · Beans and peas before leafy greens

Bacteria in the roots of beans and peas capture nitrogen from the air. Cut spent plants at the base, leaving the roots; as the remains decay, nitrogen feeds the following crop. Follow with leafy greens and brassicas as part of a rotation system — see pages 156–157.



2 · Turn plant waste into plant food

Mix green waste and peelings with shredded cardboard and woody trimmings. Keep damp, not soaked; spread on beds once dark and crumbly. Garden compost adds most micronutrients, and nitrogen, phosphorus and potassium in reasonably correct proportion, providing you add a balanced mix of ingredients.



3 · If you live near the sea, gather seaweed

After winter storms, gather loose, washed-up seaweed (where permitted). Remove plastic and spread on empty beds to decay before planting. It adds potassium, trace nutrients and organic matter. Pile it up to 6 inches / 15 cm deep — by planting season it will have rotted down into the soil and more or less disappeared.